

5. Data Models

The system defines several core entities.

Drone

Attributes:

id
name
current_position
current_height
state
assigned_route

Possible states:

idle
flying
returning
landing

Mission

Represents a drone mission.

Attributes:

mission_id
drone_id
start_point
goal_point
obstacles
status

Route

Represents the generated path.

Attributes:

route_id
drone_id
route_points
distance

Each route consists of a sequence of geographic coordinates.

Obstacle

Represents objects that block drone movement.

Attributes:

id
source
type
geometry
height

Sources:

map
user

Types:

building
restricted_zone